Reputational Bargaining and Spying

Yi Chen
Cornell Johnson

Hanzhe Zhang
Michigan State

Zekai Zhang
Harbin Tech Shenzhen

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Reputational bargaining and spying

- Bargainers build a **reputation** for being tough in **bilateral bargaining**.
  - Some bargainers are genuinely committed to their stated demands.
  - Strategic bargainers pretend to be committed.
- Bargainers **spy** to acquire information about opponents' commitments.
  - companies in mergers and acquisitions
  - countries in wars and diplomatic negotiations

Research gap and questions

- Research gap
  - Information acquisition in reputational bargaining remains underexplored.
  - We link **reputational bargaining** and **information acquisition** in a unified dynamic framework.
- Research questions
  - How does information acquisition affect bargaining behavior?
  - How does information acquisition affect bargaining power?

Model

- Bargaining (Abreu and Gul, 2000, Ecma)
  - Player i announces a fixed demand $a_i \in (0, 1)$.
    - Define conflict $D \equiv a_1 + a_2 - 1 > 0$.
  - Player i is committed to a demand a_i with probability $z_i \in (0, 1)$.
    - With probability $1 - z_i$, i is rational and flexible regarding the demand.
  - At each instant $t \geq 0$, player i decides to (i) concede or (ii) wait.
    - Player i has a discount rate r_i.
- Spying: With a Poisson rate γ_i, i receives a signal stating j is rational.
  - The signal is wrong with probability $\epsilon_i \in [0, 1/2)$.
- Parameters of the game $B = (a_i, z_i, r_i, \gamma_i, \epsilon_i)$

Abreu-Gul baseline

- $\gamma_i = 0$: players are uninformed.
- **Theorem 0 (Abreu and Gul, 2000, Ecma)**. Bargaining game $B = (a_i, z_i, r_i)$ has a unique equilibrium. There exists a time T such that the equilibrium strategy profile satisfies the following properties.

Time 0. At most one player concedes with a positive probability.

WOA. At time $t \in (0, T]$, rational i mixes between conceding and waiting.

- War of attrition between rational players
- Rational players concede by time T: $\mu_1(T) = \mu_2(T) = 1$.

Abreu-Gul concessions

- Uninformed $i = 1, 2$ is indifferent between waiting and conceding:

$$\underbrace{r_i(1 - a_j)}_{\text{waiting cost}} = \underbrace{\lambda_j(t)D}_{\text{gain from } j \text{ conceding}}$$

when j concedes at a constant overall rate

$$\lambda_j^{AG} \equiv \frac{r_i(1 - a_j)}{D} = \frac{r_i(1 - a_j)}{a_1 + a_2 - 1}.$$

  - Uninformed i concedes at a rate of

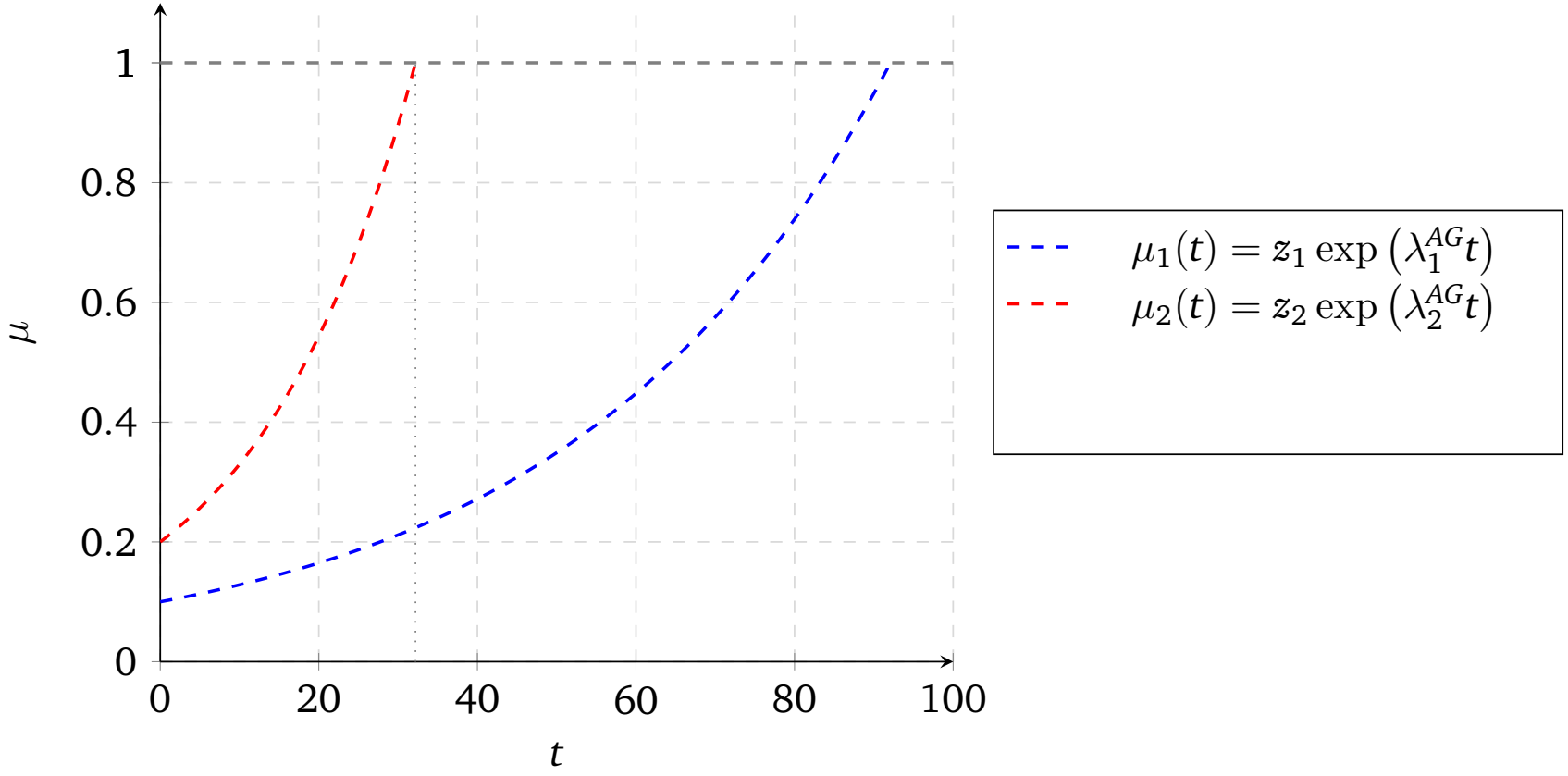
$$\frac{\lambda_i^{AG}}{1 - \mu_i(t)}.$$

Abreu-Gul reputation evolution

- Player i's reputation evolution at time t:

$$\dot{\mu}_i(t) = \lambda_i^{AG}\mu_i(t) \Longrightarrow \mu_i(t) = \mu_i(0^+)\exp\left(\lambda_i^{AG}t\right)$$

with boundary conditions $\mu_1^*(T) = \mu_2^*(T) = 1$.



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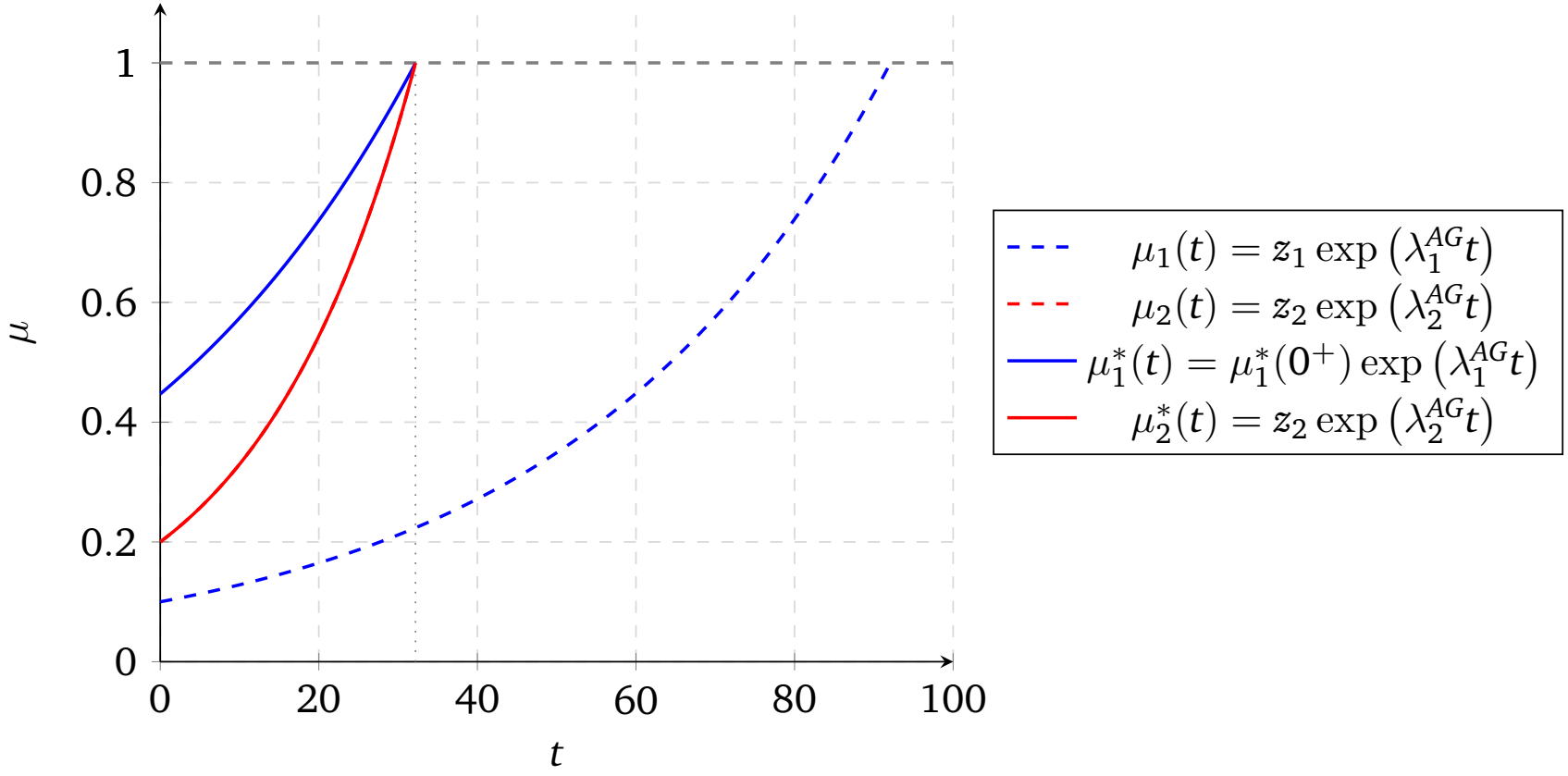
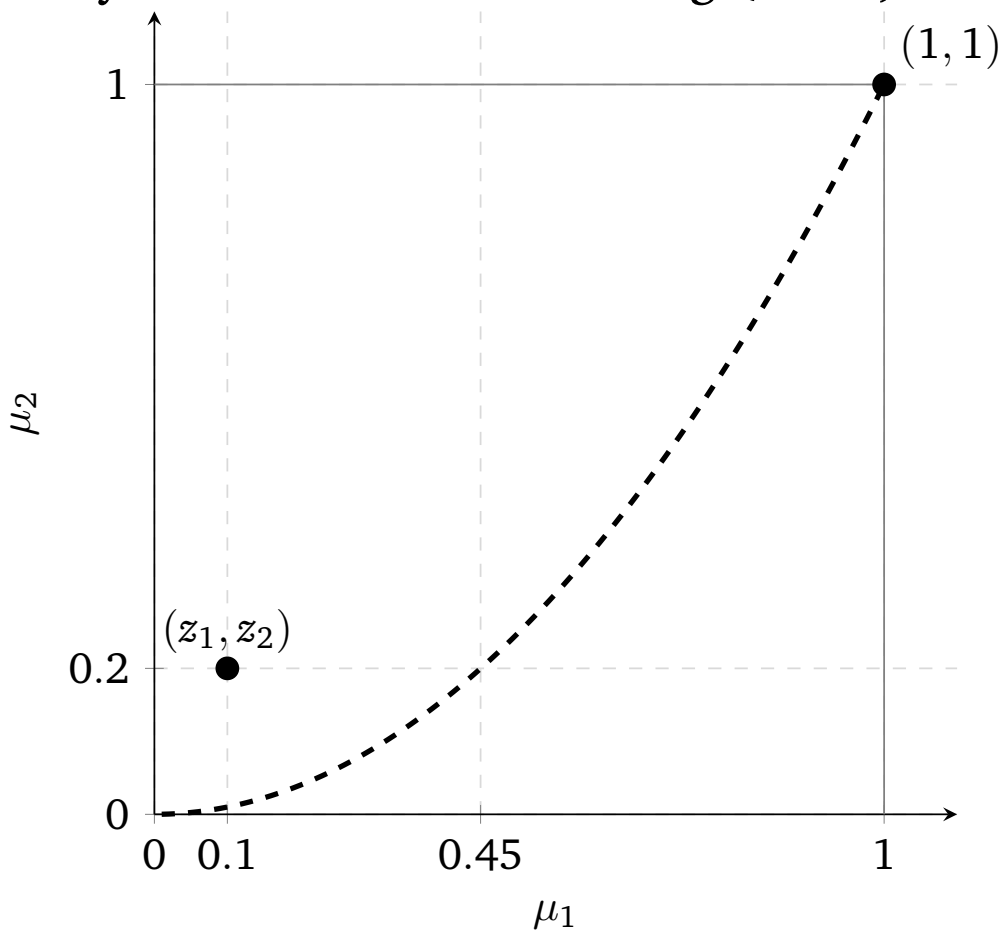


Abreu-Gul reputation coevolution

Reputation coevolution curve
in the style of Ekmekci and Zhang (2025, REStud)



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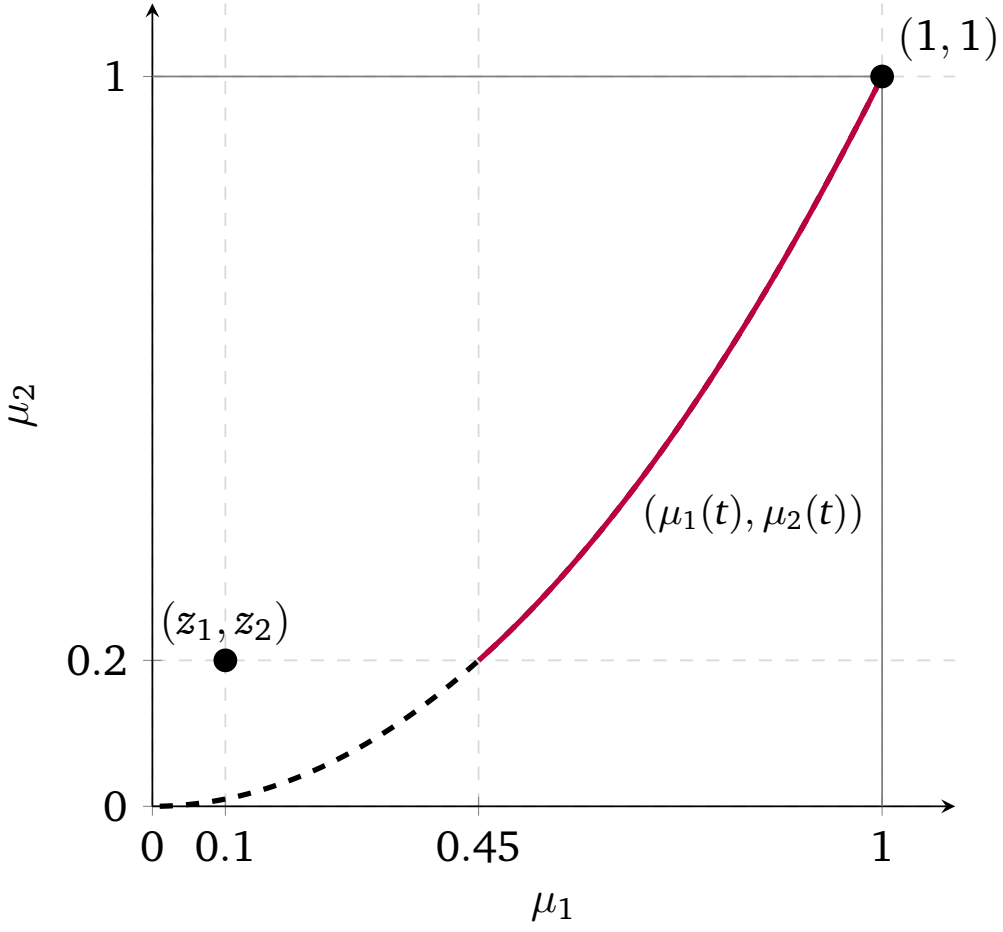

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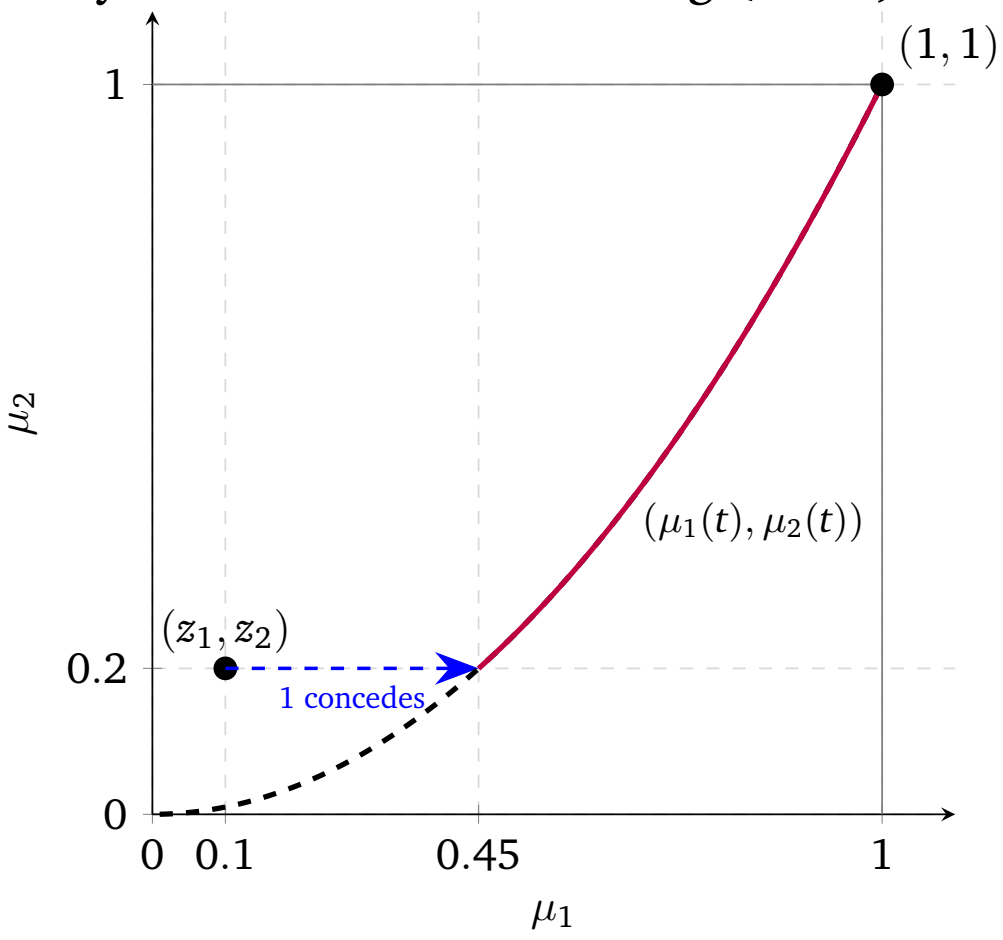


Perfect-signal baseline

- $\gamma_1 > 0$ and $\epsilon_1 = 0$.
- Rational 1 can be either informed or uninformed.
- Informed 1's continuation value $V_1(t)$.
- Value of information $\underline{V}_1(t) \equiv V_1(t) - (1 - a_2)$.
- Value of information as a fraction of conflict:

$$x(t) \equiv \frac{\underline{V}_1(t)}{D} = \frac{V_1(t) - (1 - a_2)}{a_1 - (1 - a_2)}.$$

Uninformed 1

- Uninformed 1 is indifferent between waiting and conceding:

$$\underbrace{r_1(1-a_2)}_{\text{waiting cost}} = \underbrace{\lambda_2(t)D}_{\text{gain from 2 conceding}} + \underbrace{\gamma_1[1-\mu_2(t)]\underline{V}_1(t)}_{\text{gain from spying}}.$$

- Player 2 concedes at a time-dependent overall rate that makes uninformed 1 indifferent between waiting and conceding:

$$\lambda_2(t) = \lambda_2^{AG} - \gamma_1[1-\mu_2(t)]x(t).$$

  - Rational 2 concedes at a rate

$$\frac{\lambda_2(t)}{1-\mu_2(t)}.$$

  - 2 concedes at a higher rate than in AG to keep uninformed 1 indifferent because 1 has an additional incentive to waiting.

Informed 1

- Informed 1 strictly prefers waiting to conceding:

$$\underbrace{r_1 V_1(t)}_{\text{waiting cost}} = \underbrace{\overbrace{\frac{\lambda_2(t)}{1-\mu_2(t)}}^{\text{concession rate of a rational 2}} [a_1 - V_1(t)]}_{\text{gain from 2 conceding}} + \dot{V}_1(t)$$

  - Plug in $\underline{V}_1(t)$:

$$r_1[\underline{V}_1(t) + (1-a_2)] = \frac{\lambda_2(t)}{1-\mu_2(t)}[D - \underline{V}_1(t)] + \dot{\underline{V}}_1(t).$$

    Divide by D:

$$\dot{x}(t) = r_1 x(t) + \lambda_2^{AG} - \frac{\lambda_2(t)}{1-\mu_2(t)}(1-x(t)).$$

    Plug in $\lambda_2(t)$ and rearrange:

$$\dot{x}(t) = [r_1 + \gamma_1(1-x(t))]\,x(t) + \lambda_2^{AG}\frac{x(t)-\mu_2(t)}{1-\mu_2(t)}.$$

Rational 2

- Rational 2 is indifferent between waiting and conceding:

$$\underbrace{r_2(1-a_1)}_{\text{waiting cost}} = \underbrace{\lambda_1(t)D}_{\text{waiting gain from 1 conceding}}.$$

- 1 concedes at a constant overall concession rate that makes rational 2 indifferent between waiting and conceding:

$$\lambda_1(t) = \frac{r_2(1-a_1)}{D} = \lambda_1^{AG}.$$

2's reputation

- 2's reputation evolution (from uninformed 1's perspective) is

$$\dot{\mu}_2(t) = \mu_2(t)\lambda_2(t).$$

- Plug in $\lambda_2(t)$:

$$\frac{\dot{\mu}_2(t)}{\mu_2(t)} = \lambda_2^{AG} - \gamma_1[1 - \mu_2(t)]x(t).$$

1's reputation

- 1's no-concession reputation μ_1^N is 1's probability of not conceding.
- 1's no-concession reputation evolution is

$$\frac{\dot{\mu}_1^N(t)}{\mu_1^N(t)} = \frac{\lambda_1^{AG}}{1-\mu_1^N(t)} + \gamma_1.$$

- 1's commitment-type reputation evolution is

$$\frac{\dot{\mu}_1(t)}{\mu_1(t)} = \frac{\lambda_1^{AG}}{1-\mu_1^N(t)}.$$

System of differential equations, $t \geq 0$

- 1's value:

$$\dot{x}(t) = [r_1 + \gamma_1(1 - x(t))]\, x(t) + \lambda_2^{AG} \left[1 - \frac{1 - x(t)}{1 - \mu_2(t)}\right],$$

  with $x(T) = 0$: signal is strictly valuable until the last instant.

- 2's reputation:

$$\frac{\dot{\mu}_2(t)}{\mu_2(t)} = \lambda_2^{AG} - \gamma_1[1 - \mu_2(t)]x(t),$$

  with $\mu_2(T) = 1$.

- 1's no-concession reputation:

$$\frac{\dot{\mu}_1^N(t)}{\mu_1^N(t)} = \frac{\lambda_1^{AG}}{1 - \mu_1^N(t)} + \gamma_1,$$

  with $\mu_1^N(T) = 1$.

System of differential equations backward, $\tau \leq 0$

- 1's value:

$$\dot{x}(\tau) = [r_1 + \gamma_1(1 - x(\tau))]\, x(\tau) + \lambda_2^{AG} \left[1 - \frac{1 - x(\tau)}{1 - \mu_2(\tau)}\right],$$

  with $x(0) = 0$.

- 2's reputation:

$$\frac{\dot{\mu}_2(\tau)}{\mu_2(\tau)} = \lambda_2^{AG} - \gamma_1[1 - \mu_2(\tau)]x(\tau),$$

  with $\mu_2(0) = 1$.

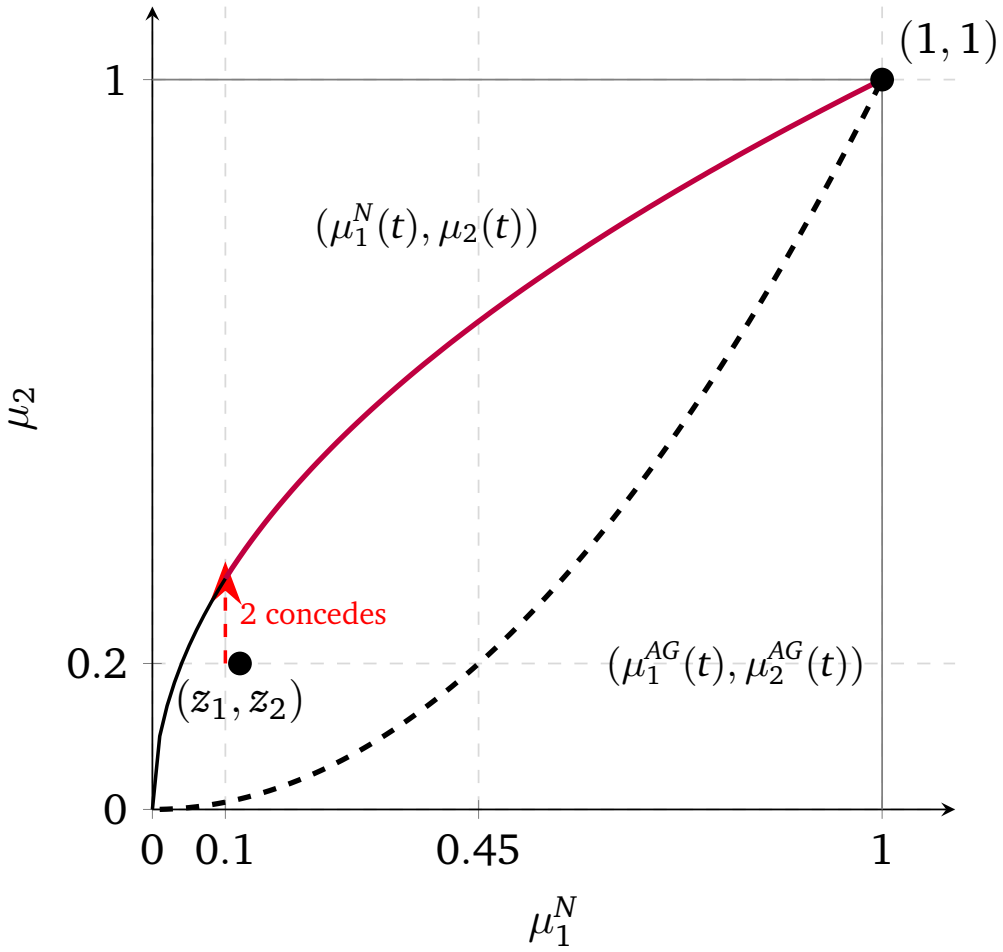
- 1's no-concession reputation:

$$\frac{\dot{\mu}_1^N(\tau)}{\mu_1^N(\tau)} = \frac{\lambda_1^{AG}}{1 - \mu_1^N(\tau)} + \gamma_1,$$

  with $\mu_1^N(0) = 1$.

Reputation coevolution curve

- Reputation coevolution $(\mu_1^N(t), \mu_2(t))$ with $\mu_1^N(T) = \mu_2(T) = 1$.



Noisy-signal model

- $\gamma_1 > 0$ and $\epsilon_1 \in (0, 1/2)$: player 1 can receive a noisy signal.
- Two phases:

Time 0. Mass concession: At most one player concedes at time 0.

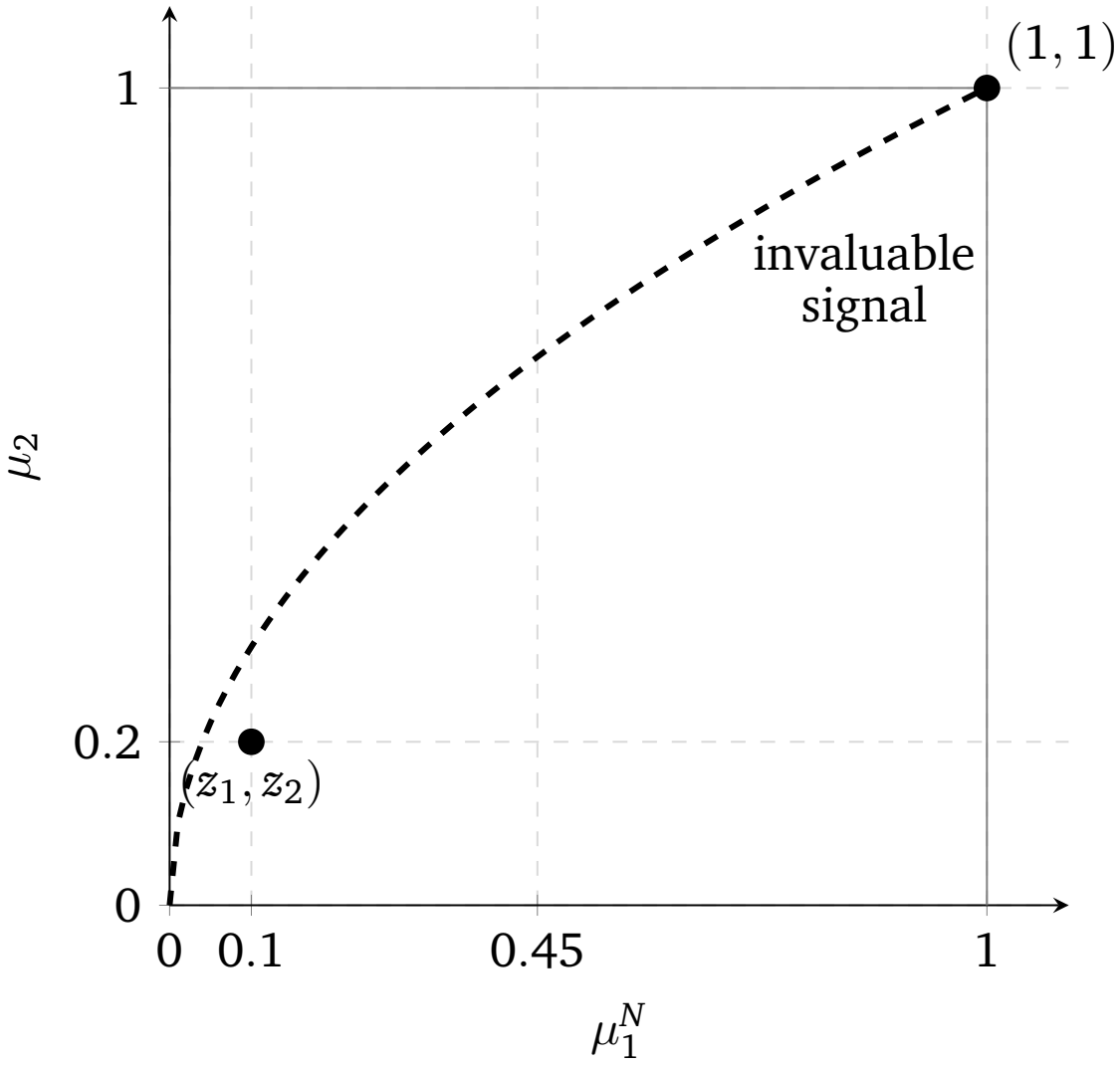
Phase 1. Valuable-signal phase: WOA between uninformed 1 and rational 2

- Uninformed 1 waits for the signal to arrive (while conceding).
- 2 concedes at a variable, higher-than-AG rate that makes uninformed 1 indifferent between waiting and conceding.
- Informed 1 strictly prefers waiting to conceding.

Phase 2. Invaluable-signal phase: WOA between informed 1 and rational 2.

- Uninformed 1 has either become informed or conceded.
- 2 concedes at the constant overall AG rate that makes informed 1 indifferent between waiting and conceding.

Reputation coevolution curve



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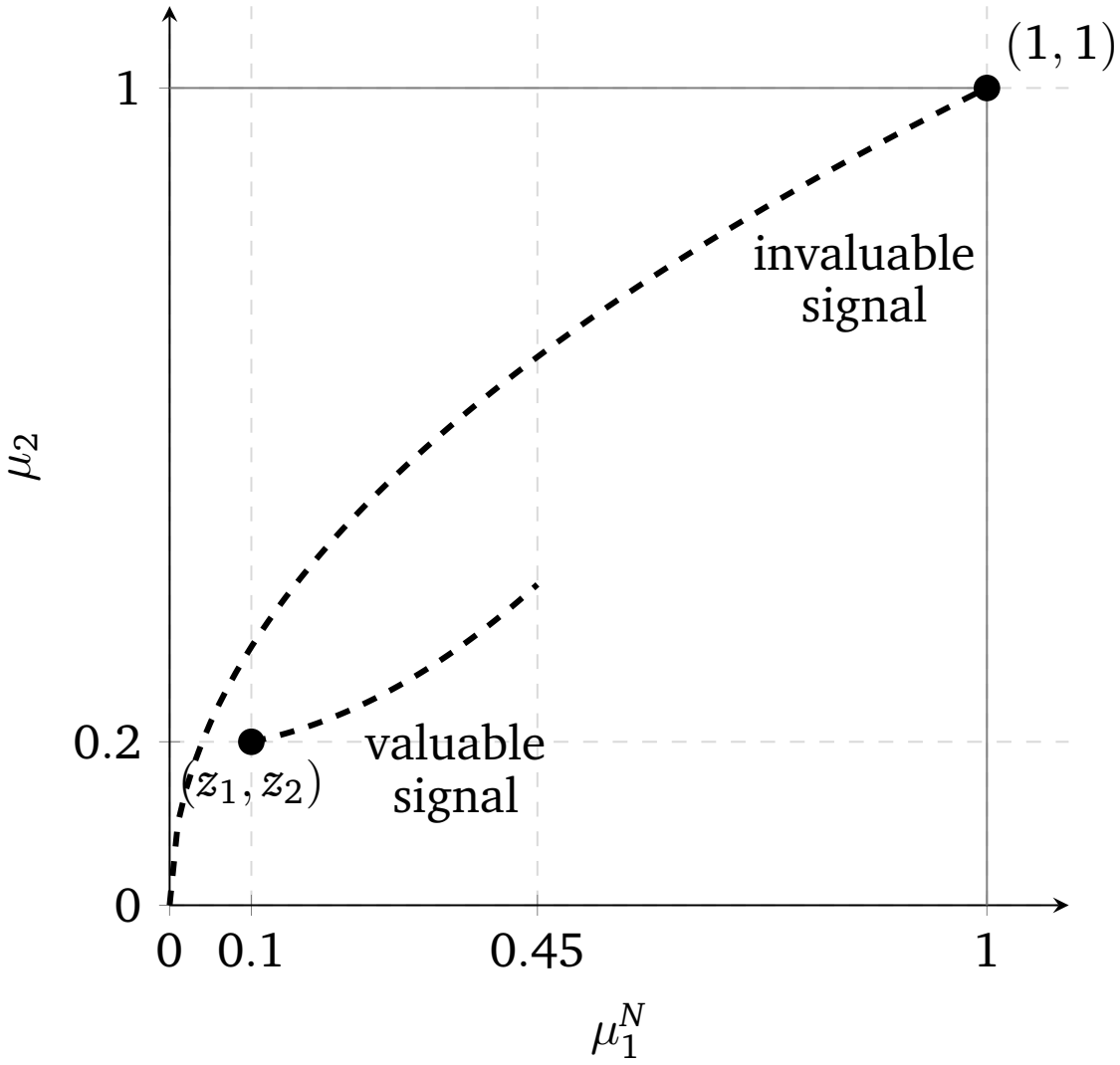

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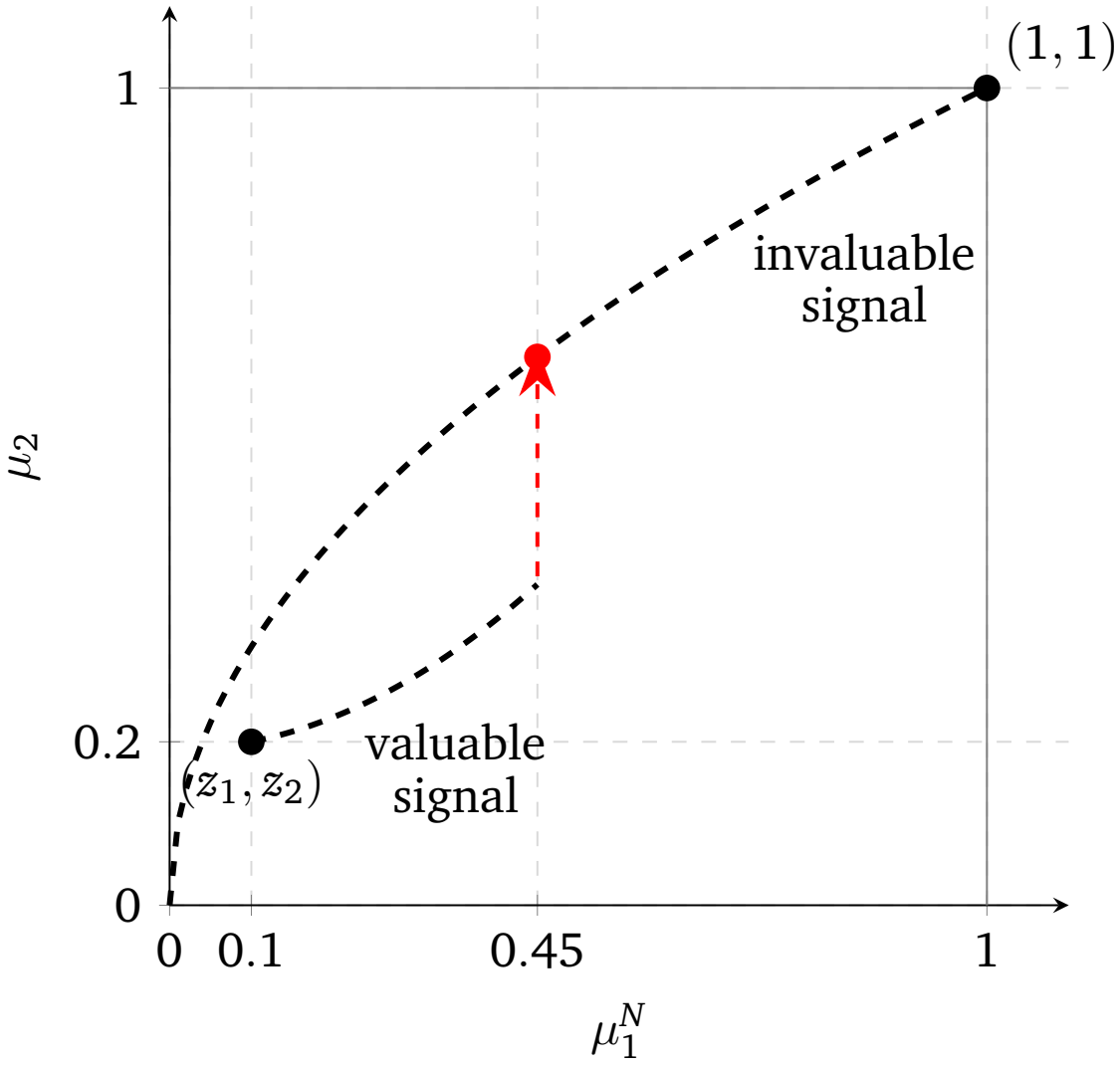

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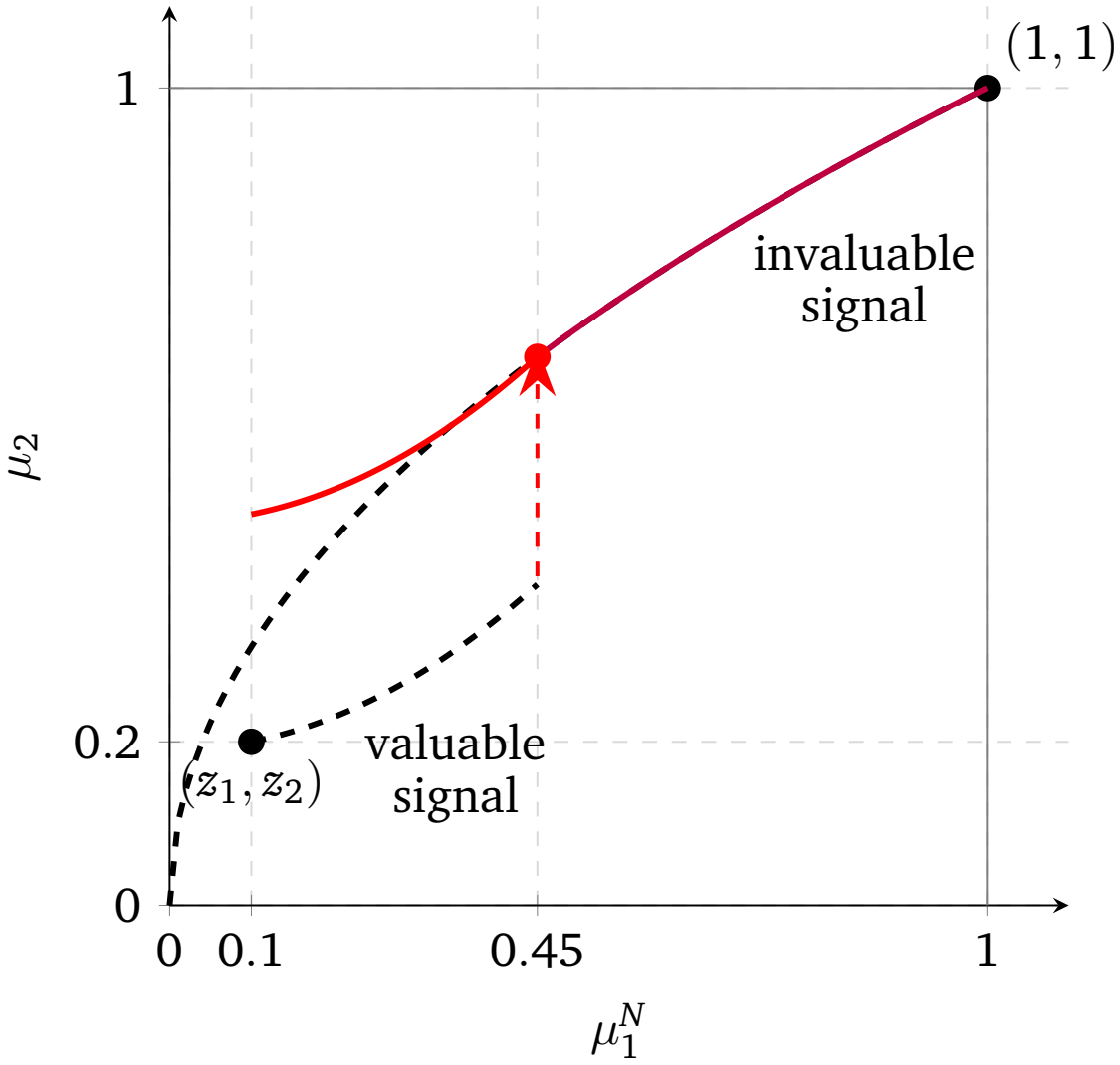

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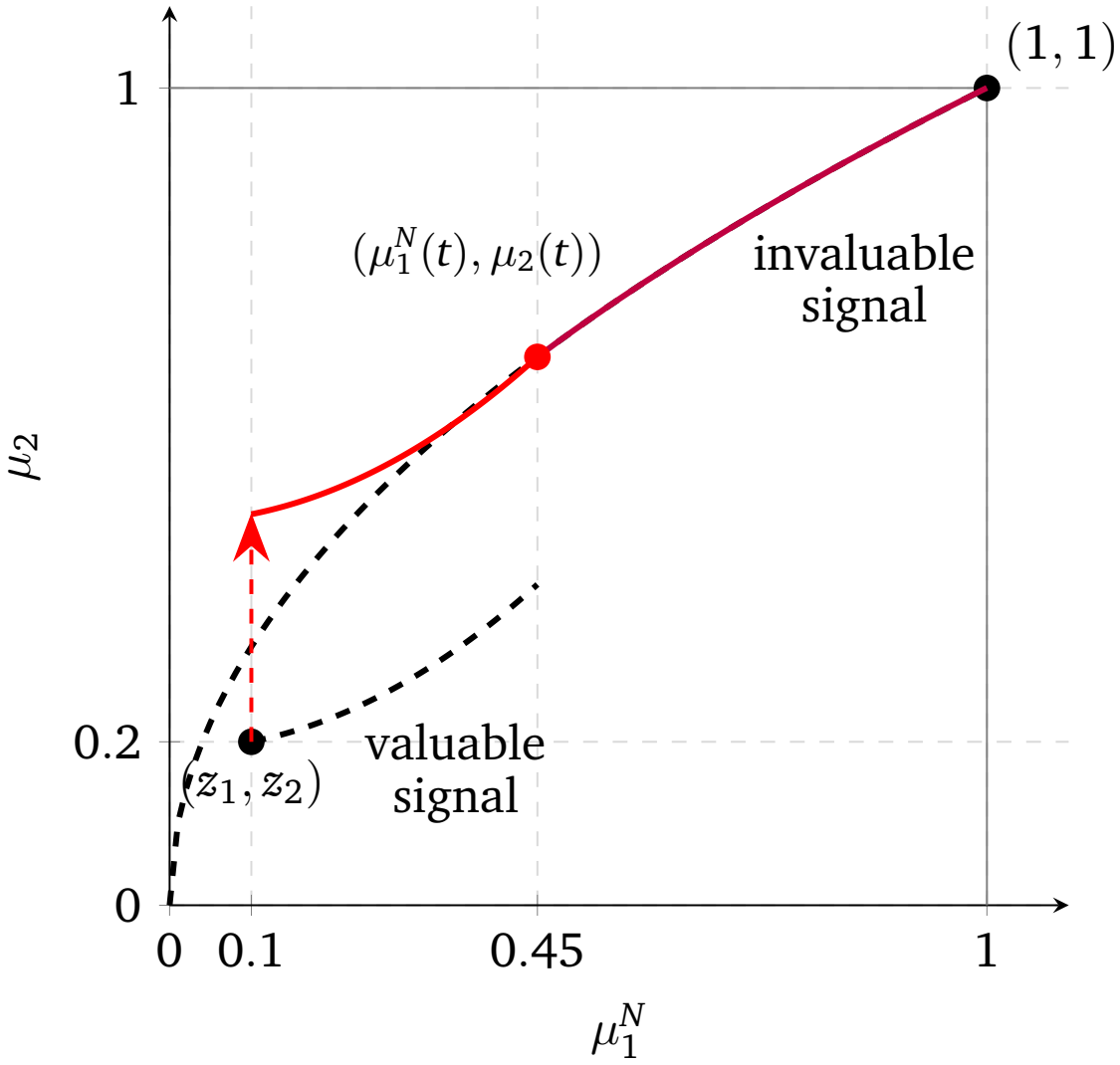


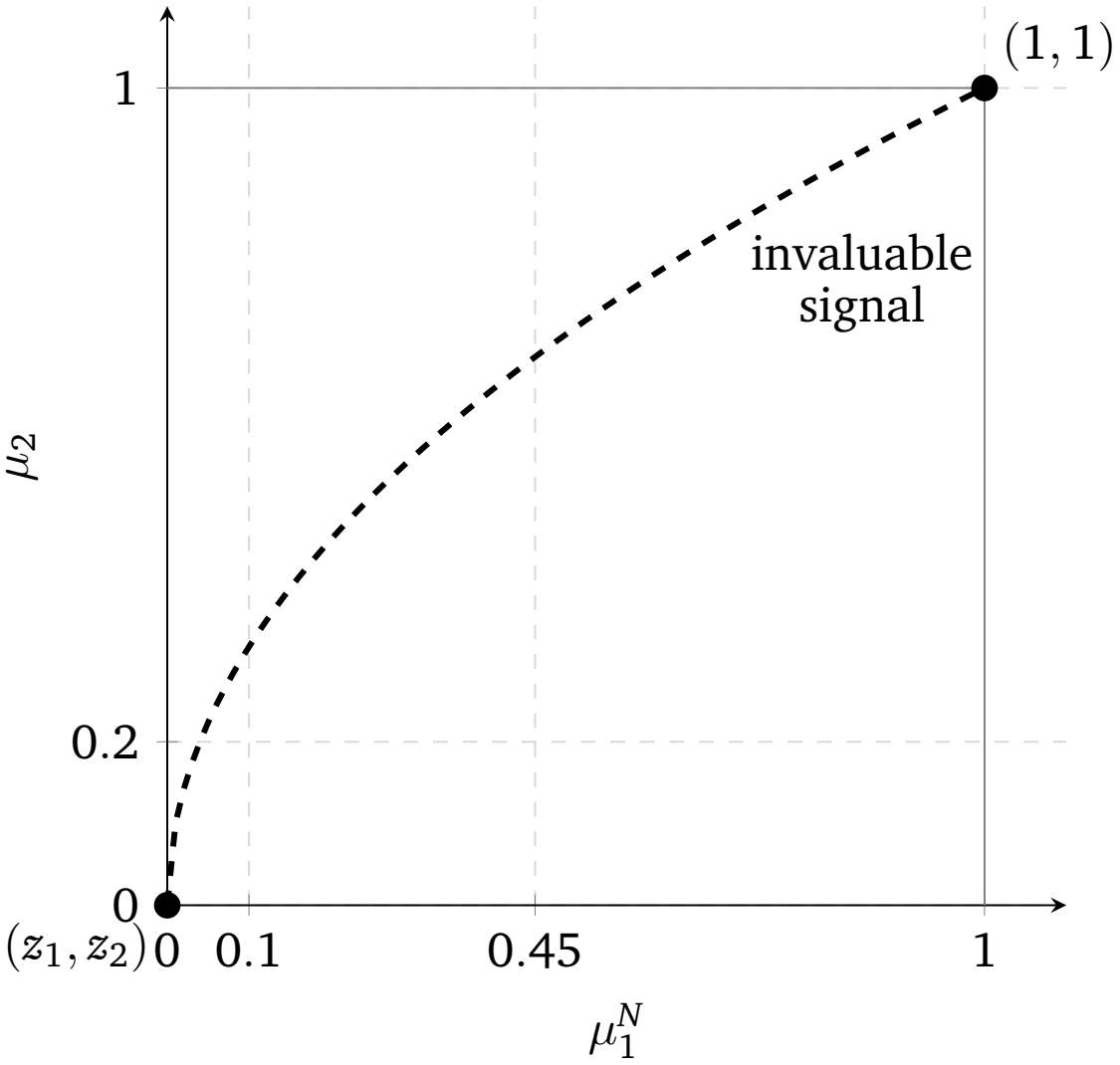
Coase Conjecture with two-sided private information

Proposition 1 (Coase Conjecture)
Let $\{B^n = (z_1^n, z_2^n, a_1, a_2, r_1, r_2, \gamma_1, \gamma_2 = 0)\}_n$ be a sequence of single-demand bargaining games with unilateral exogenous spying. Let u_i^n be the equilibrium payoff of rational Player i in B^n. If $\lim_{n\to\infty}(z_1^n, z_2^n) = 0$,

$$\lim_{n\to\infty} u_1^n = a_1 \text{ and } \lim_{n\to\infty} u_2^n = 1 - a_1.$$

- Gul et al. (1986, JET): unilateral incomplete information implies immediate agreements favorable to the informed.
- Abreu and Gul (2000, Ecma): in two-sided private information, relative impatience plays a key role.
- Our paper restores the Coase conjecture result under two-sided private information with unilateral information acquisition.

Reputation coevolution curve



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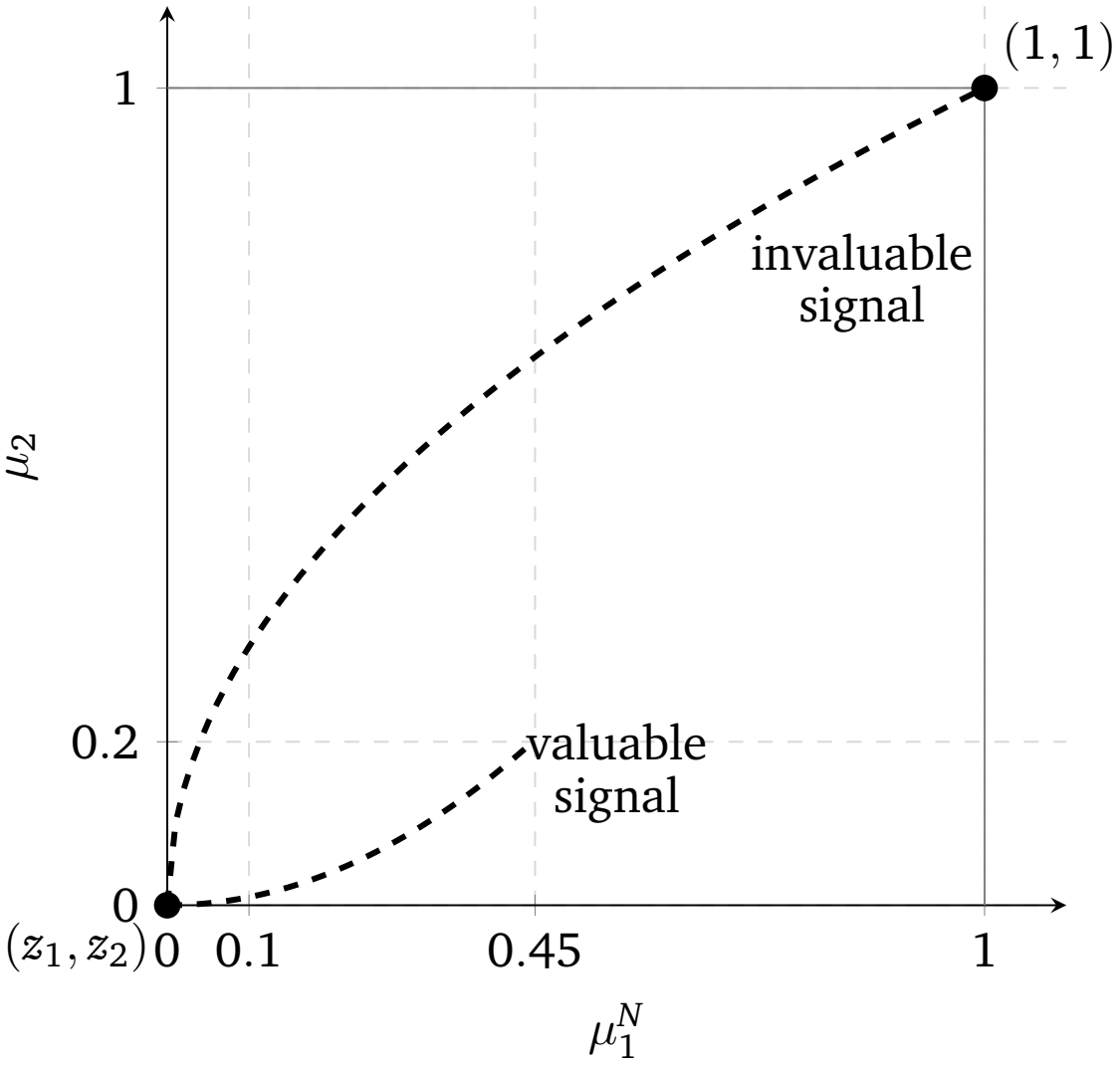

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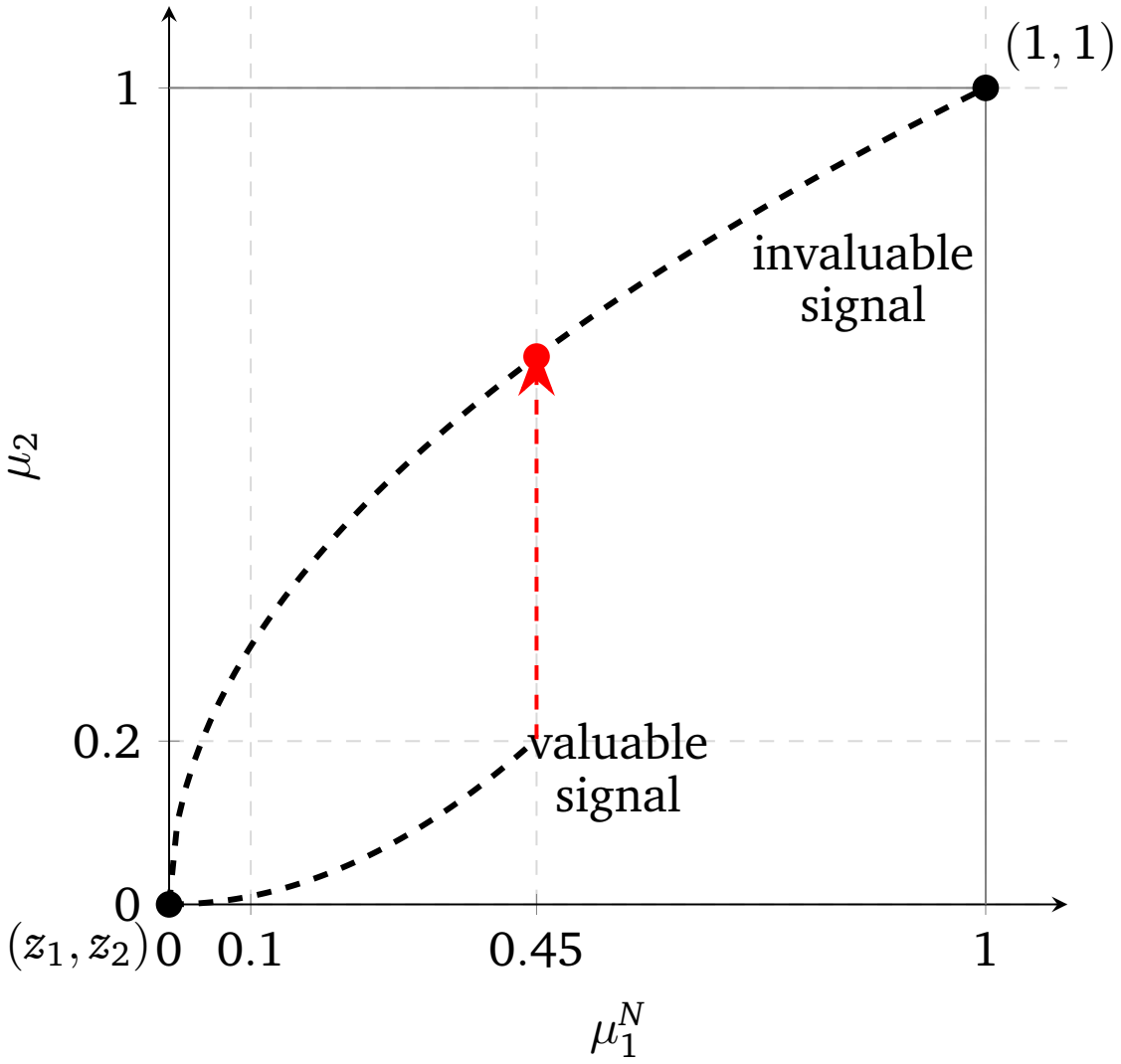

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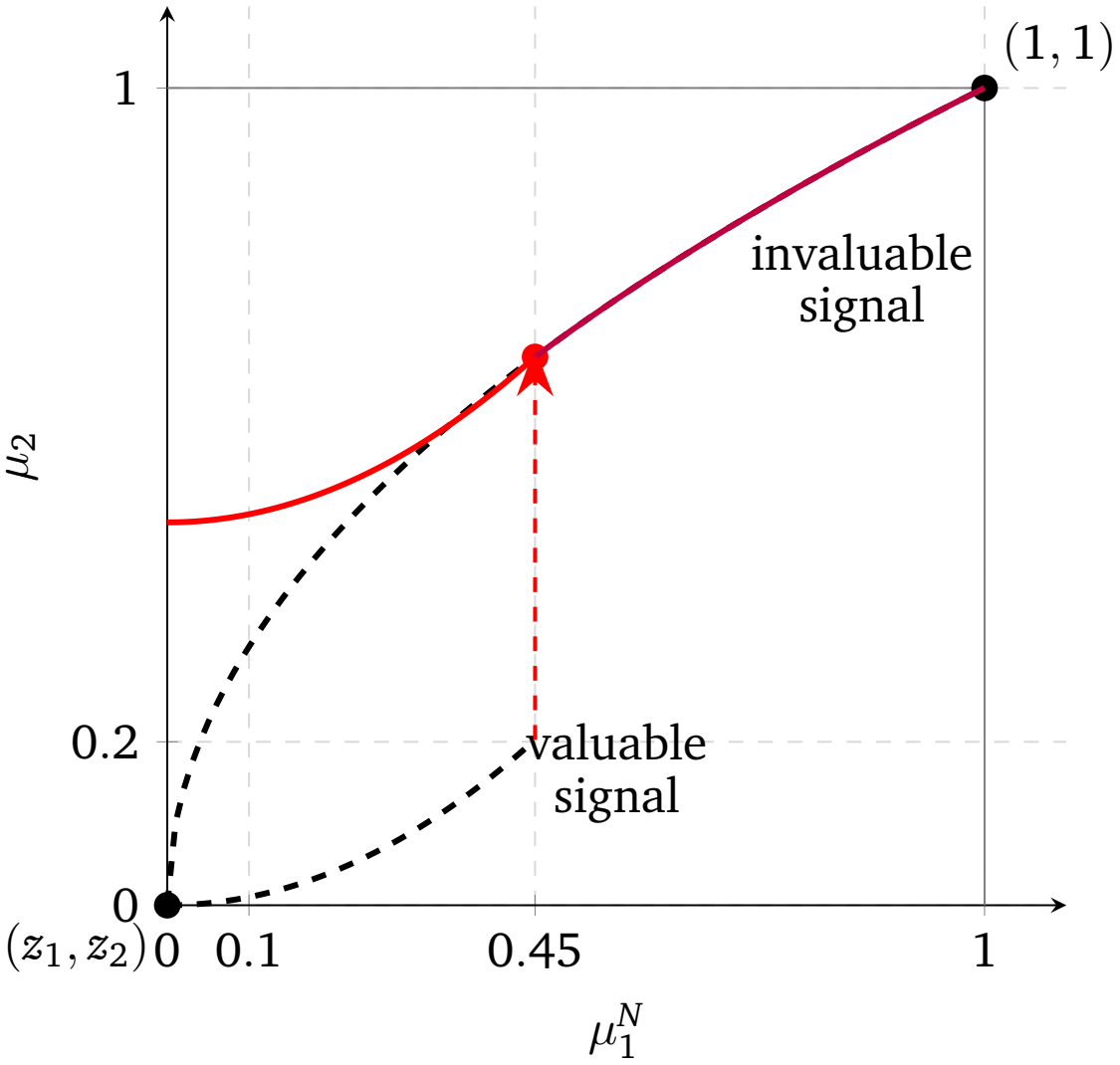

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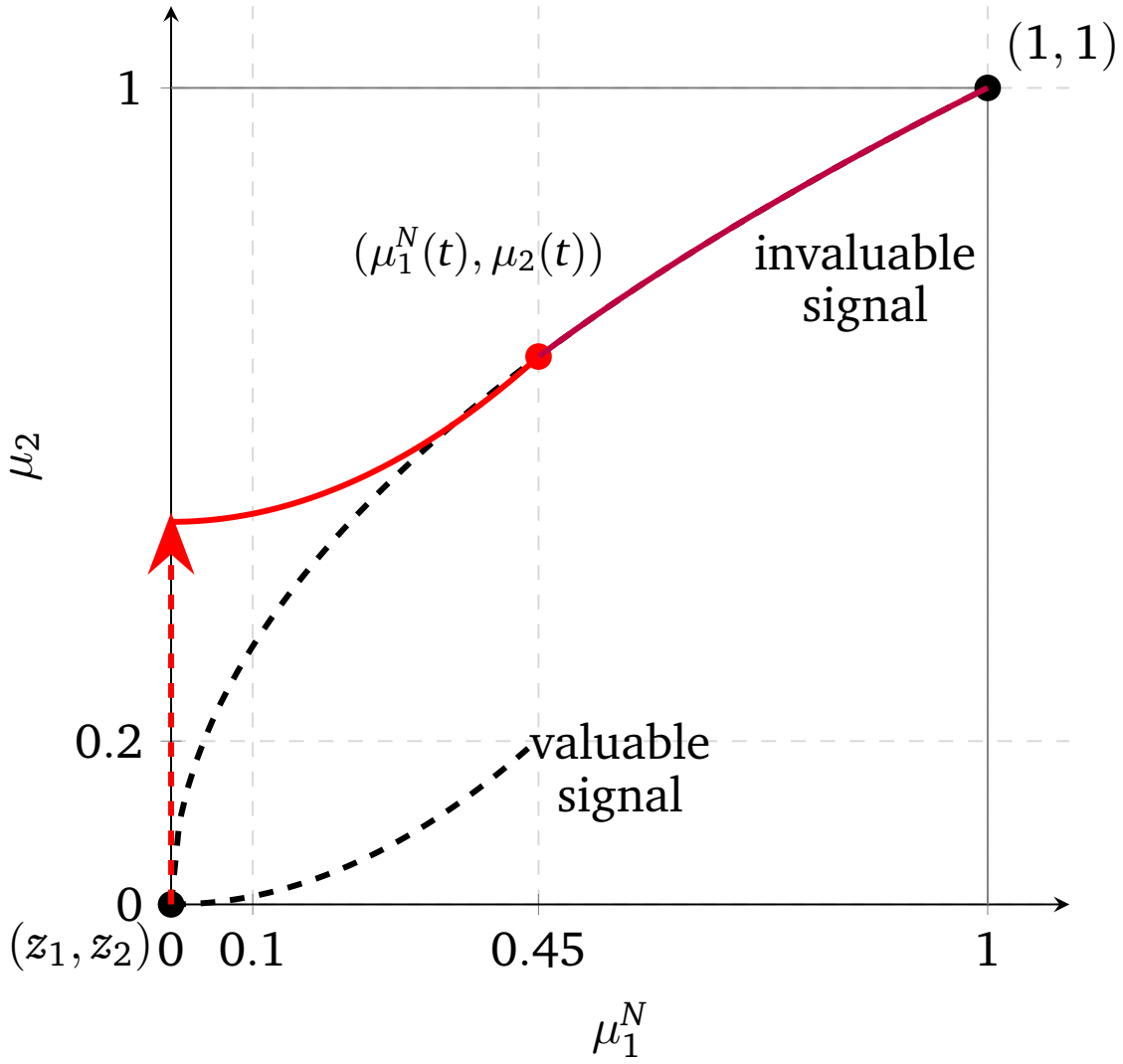

Two-sided noisy-signal model

- Suppose $\gamma_1 > 0$ and $\gamma_2 > 0$.
- Generically, there are three phases in equilibrium.

Time 0. Mass concession: At most one player concedes at time 0.

Phase 1. Double-valuable signal: WOA between the uninformed

- The signal is valuable for informed 1 and informed 2.

Phase 2. Single-valuable signal: WOA between informed i and uninformed j

- i is surely informed, and j may be uninformed.
- The signal is strictly valuable for informed j.

Phase 3. Invaluable signal: WOA between the informed

- Both players are surely informed but concede with positive probabilities.
- The signal is no longer valuable for either informed player.

Conclusion

- We add spying opportunities to the reputational bargaining framework of Abreu and Gul (2000, Ecma).

  We provide a result in the spirit of Coase Conjecture in two-sided private information.

- We fully characterize the unique equilibrium.

- It remains to explore the results regarding comparative statics, multiple demand types, and limit results, as is often explored by the literature.

THANK YOU!

Uninformed 1 in the spying phase

- Uninformed 1 is indifferent between waiting and conceding:

$$\underbrace{r_1(1-a_2)}_{\text{waiting cost}} = \underbrace{\lambda_2(t)D}_{\text{gain from 2 conceding}} + \underbrace{\gamma_1[1-\mu_2(t)]\underline{V}_1(t)}_{\text{spying gain}}.$$

- Player 2's overall concession rate is

$$\implies \lambda_2(t) = \lambda_2^{AG} - \gamma_1[1-\mu_2(t)]x(t).$$

Informed 1 in the spying phase

- Informed 1's posterior belief about 2 after a signal:

$$\nu_2(t) = \frac{\mu_2(t)\epsilon_1}{\mu_2(t)\epsilon_1 + [1-\mu_2(t)][1-\epsilon_1]}.$$

- Informed 1 strictly prefers waiting to conceding in the spying phase:

$$r_1 V_1(t) = [1-\nu_2(t)]\frac{\lambda_2(t)}{1-\mu_2(t)}[a_1 - V_1(t)] + \dot{V}_1(t).$$

- Simplified,

$$\dot{x}(t) = [r_1 + \gamma_1(1-\nu_2(t))(1-x(t))]\,x(t) + \lambda_2^{AG}\left[1 - \frac{(1-\nu_2(t))(1-x(t))}{1-\mu_2(t)}\right].$$

2's reputation in the spying phase

- 2's reputation when 1 is uninformed:

$$\dot{\mu}_2(t) = \mu_2(t)\lambda_2(t).$$

- 2's reputation evolution in the spying phase:

$$\implies \frac{\dot{\mu}_2(t)}{\mu_2(t)} = \lambda_2^{AG} - \gamma_1[1 - \mu_2(t)]x(t).$$

- By Bayes Rule,

$$\frac{1 - \nu_2(t)}{\nu_2(t)} = \frac{1 - \epsilon_1}{\epsilon_1} \cdot \frac{1 - \mu_2(t)}{\mu_2(t)},$$

then (by a lot of algebra),

$$\frac{1 - \nu_2(t)}{1 - \mu_2(t)} = 1 + \nu_2(t)\frac{1 - 2\epsilon_1}{\epsilon_1}.$$

System of differential equations

- Define signal precision

$$\rho_1 = \frac{1 - 2\epsilon_1}{\epsilon_1} \in [0, \infty), \forall \epsilon_1 \in (0, 1/2].$$

- Then

$$\dot{\nu}_2(t) = \nu_2(t) \left[\lambda_2^{AG}(1 + \rho_1 \nu_2(t)) - \gamma_1(1 - \nu_2(t))x(t) \right].$$

$$\dot{x}(t) = r_1 x(t) + \lambda_2^{AG} - (1 - x(t)) \left[\lambda_2^{AG}(1 + \rho_1 \nu_2(t)) - \gamma_1(1 - \nu_2(t))x(t) \right].$$

R2 in the spying phase

- Rational 2 is indifferent between waiting and conceding:

$$\underbrace{r_2(1-a_1)}_{\text{waiting cost}} = \underbrace{\lambda_1(t)D}_{\text{waiting gain from Player 1 conceding}}.$$

- 1 concedes at a constant overall rate in the doubt phase:

$$\lambda_1(t) = \frac{r_2(1-a_1)}{D} = \lambda_1^{AG}.$$

- Uninformed 1 concedes at a rate

$$\lambda_1^{\emptyset}(t) = \frac{\lambda_1^{AG}}{1-\mu_1^N(t)}.$$

Reputation evolution of being uninformed

- The probability of 1 being uninformed is $\mu_1^{\emptyset}(t) = 1 - \mu_1^{N}(t)$.

$$\dot{\mu}_1^{\emptyset}(t) = -\lambda_1^{AG}(1 - \mu_1^{\emptyset}(t)) - \gamma_1 \mu_1^{\emptyset}(t).$$

- The boundary condition of the phase is

$$\mu_1^{\emptyset}(T_s) = 0.$$

- Given the initial reputation $\mu_1^{\emptyset}(0^+) = 1 - \mu_1(0^+)$, the length of the spying phase is

$$T_s = \frac{1}{\lambda_1^{AG} - \gamma_1} \ln \left[\frac{\lambda_1^{AG}}{\lambda_1^{AG} - (\lambda_1^{AG} - \gamma_1)\mu_1^{\emptyset}(0^+)} \right].$$

- Player 1's reputation

$$\dot{\mu}_1(t) = \lambda_1^{AG} \mu_1(t).$$

I1 in the doubt phase

- Informed 1 is indifferent between waiting and conceding:

$$V_1(t) = 1 - a_2.$$

- HJB:

$$\underbrace{r_1(1 - a_2)}_{\text{waiting cost}} = \underbrace{\lambda_2(t)D}_{\text{waiting gain from Player 2 conceding}}.$$

- Player 2's overall concession rate in the doubt phase:

$$\implies \lambda_2(t) = \frac{r_1(1 - a_2)}{D} = \lambda_2^{AG}.$$

R2 in the doubt phase

- Rational 2 is indifferent between waiting and conceding:

$$\underbrace{r_2(1-a_1)}_{\text{waiting cost}} = \underbrace{\lambda_1(t)D}_{\text{waiting gain from Player 1 conceding}}.$$

- Player 1's overall concession rate in the doubt phase:

$$\implies \lambda_1(t) = \frac{r_2(1-a_1)}{D} = \lambda_1^{AG}.$$

Reputation coevolution in the doubt phase

- For both players, the only news is that their opponent has not conceded.
- Player 1's reputation:

$$\dot{\mu}_1(t) = \lambda_1(t)\mu_1(t) \Longrightarrow \frac{\dot{\mu}_1(t)}{\mu_1(t)} = \lambda_1^{AG},$$

with $\mu_1(T) = 1$.

- Player 2's reputation to an informed 1:

$$\dot{\nu}_2(t) = \lambda_2(t)\nu_2(t) \Longrightarrow \frac{\dot{\nu}_2(t)}{\nu_2(t)} = \lambda_2^{AG},$$

with $\nu_2(T) = 1$.

Reputation coevolution in the spying phase

- Start without a mass concession: $\mu_1(0) = z_1$ and $\mu_2(0) = z_2$.
  - The initial belief of informed 1 is

$$\nu_2(0) = \frac{z_2\epsilon_1}{z_2\epsilon_1 + (1-z_2)(1-\epsilon_1)}.$$

- Spying phase:

$$\dot{\nu}_2(t) = \nu_2(t)\left[\lambda_2^{AG}(1+\rho_1\nu_2(t)) - \gamma_1(1-\nu_2(t))x(t)\right].$$

$$\dot{x}(t) = r_1x(t) + \lambda_2^{AG} - (1-x(t))\left[\lambda_2^{AG}(1+\rho_1\nu_2(t)) - \gamma_1(1-\nu_2(t))x(t)\right].$$

- Given $\mu_1^{\emptyset}(0^+) = 1 - \mu_1(0^+)$, the length of the spying phase is

$$T_s = \frac{1}{\lambda_1^{AG} - \gamma_1}\ln\left[\frac{\lambda_1^{AG}}{\lambda_1^{AG} - (\lambda_1^{AG} - \gamma_1)\mu_1^{\emptyset}(0^+)}\right].$$

- Player 1's reputation in the spying phase

$$\dot{\mu}_1(t) = \lambda_1^{AG}\mu_1(t) \Rightarrow \mu_1(T_s) = \mu_1(0^+)\exp(\lambda_1^{AG}T_s).$$

Boundary conditions

- Start from $(\mu_1(0) = z_1, \nu_2(0) = \frac{z_2\epsilon_1}{z_2\epsilon_1+(1-z_2)(1-\epsilon_1)})$.
- Calculate $T_s(1-z_1) = \frac{1}{\lambda_1^{AG}-\gamma_1} \ln\left[\frac{\lambda_1^{AG}}{\lambda_1^{AG}-(\lambda_1^{AG}-\gamma_1)(1-z_1)}\right]$.
- Find $\nu_2(T_s(1-z_1))$ from $\dot{x}$, $\dot{\nu}_2$, and $x(T_s) = 0$, and find $\mu_1(T_s) = \mu_1(0^+)\exp(\lambda_1^{AG}T_s)$.
- In the doubt phase, trace back the curve from $(1,1)$ to $(\mu_1(T_s), \nu_2(\mu_1(T_s)))$.

War of attrition

- Both players are now informed for sure.
- Let $\nu_i(t)$ denote player i's reputation to an informed j.
- Informed $i = 1, 2$ concedes at a rate of

$$\lambda_i^{AG} / \left(1 - \nu_i^{AG}(t)\right).$$

- Reputation evolves according to

$$\dot{\nu}_i(t) = \lambda_i^{AG} \nu_i(t).$$

Bilateral spying phase

- Uninformed i is indifferent between conceding and waiting.
  - Let μ_i denote the uniformed j's belief that i is committed.
  - Let μ_i^N denote the uniformed j's belief that i does not concede.
- Informed i strictly prefers waiting to conceding.
  - Let ν_i denote the informed j's belief that i is committed.
  - Let ν_i^N denote the informed j's belief that i does not concede.

Uninformed i in the bilateral spying phase

- Uninformed i is indifferent between conceding and waiting:

$$r_i(1 - a_j) = \lambda_j(t)D + \gamma_i[1 - \mu_j(t)]\underline{V}_i(t) \Rightarrow \lambda_j(t) = \lambda_i^{AG} - \gamma_i[1 - \mu_j(t)]x_i(t).$$

Informed i in the bilateral spying phase

- Informed i's posterior belief about j after receiving a signal:

$$\nu_j(t) = \frac{\mu_j(t)\epsilon_i}{\mu_j(t)\epsilon_i + (1-\mu_j(t))(1-\epsilon_i)}.$$

- Informed i strictly prefers waiting to conceding:

$$r_i V_i(t) = [1-\nu_j(t)]\frac{\lambda_j(t)}{1-\mu_j(t)}[a_i - V_i(t)] + \dot{V}_i(t).$$

Simplified,

$$\begin{aligned}\dot{x}_i(t) &= [r_i + \gamma_i(1-\nu_j(t))(1-x_i(t))]x_i(t) \\ &\quad + \lambda_j^{AG}\left[1 - \frac{(1-\nu_j(t))(1-x_i(t))}{1-\mu_j(t)}\right].\end{aligned}$$

References I

Abreu, D. and Gul, F. (2000). Bargaining and reputation. *Econometrica*, 68(1):85–117.

Ekmekci, M. and Zhang, H. (2025). Reputational bargaining with external resolution opportunities. *The Review of Economic Studies*, 92(4):2472–2501.

Gul, F., Sonnenschein, H., and Wilson, R. (1986). Foundations of dynamic monopoly and the coase conjecture. *Journal of Economic Theory*, 39(1):155–190.